

3. Retaining multiple copies of data sets at regular intervals provides the flexibility and insurance to restore data at any instance of time, without being affected by malicious attacks or data corruption.
4. Open source backup solutions are known for better hardware and software compatibility.
5. As these solutions are available as open source, they can be customised and configured as per the requirement.
6. They have better and wider community support, which helps in getting bugs and vulnerabilities fixed quickly.
7. The security of any open source solution is generally underestimated; in reality, it is often more secure than many licensed backup solutions.
8. Open source backup solutions also provide a better user experience, which helps any one easily use the features they require.

Different types of recovery and backup solutions

The two different approaches to recovery and backup are listed below.

Traditional or streaming backups: This is the traditional approach to recovery. All the available data is streamed from the application server or hosted through a backup server to a secondary storage system. The backup server basically ingests the data and then performs additional actions to optimise it before writing the data to the secondary medium. The additional actions commonly include:

- a. Indexing the data set for easy restore and search
 - b. Data reduction (which includes de-duplication, compression, etc)
 - c. Encrypting the available data to protect it during transit
- Traditional backup and recovery systems offer a few benefits, including:

- a. The ability to manage and consolidate recovery and backup from multiple primary systems with the help of just a single backup interface and storage target
- b. The ability to reduce the data footprint with compression and global de-duplication
- c. Application integration to improve the state of data when restored
- d. Intelligent metadata management to improve the data recovery

Array-based backup or recovery: Array-based backup and recovery solutions are built using storage snapshots and offer an alternative approach to protecting the data set. There are various benefits to this approach:

- a. High-performance snapshots on the primary storage easily create local recovery points with a very low impact on the production workload. This enables higher service levels with shorter backup windows and frequent recovery points.
- b. Local snapshots generally offer rapid recovery as compared to the streaming backup since the snapshot is already present on the primary storage and doesn't need to



Figure 1: A typical Windows backup solution (image source: [googleimages.com](https://www.googleimages.com))

- be retrieved from the secondary media.
- c. Array-based data replication removes the requirement to stream data through a virtual host or the application server.
- d. Data efficiencies are retained from the primary storage to the secondary storage.
- e. Data transferred to the secondary storage is not present in a proprietary format, hence making it faster and easier to use for other purposes like instant tests or dev workflows.

Top open source backup solutions

AMANDA

AMANDA, also known as Advanced Maryland Automatic Network Disk Archiver, is a popular enterprise backup system. It was developed at the University of Maryland in 1991 to protect files on a large number of client workstations using a single backup server. James Da Silva was one of its original developers. AMANDA runs on almost a million servers and desktops across the globe, supporting multiple operating systems. It is available in the market in three different editions—Enterprise, Community and Zmanda Backup Appliance. The community edition is available free of cost whereas the enterprise edition supports live application backups. The Zmanda backup appliance is basically a virtual machine that's capable of easily backing up an entire network.

Features:

1. Supports multiple operating systems including Windows, UNIX, BSD, MacOS, etc.
2. Supports tape, disk and optical media backups.
3. Allows users to set up a single master backup server to multiple Windows, Linux and UNIX hosts.
4. It provides backup of sparse files and hard links as well.
5. There is no change to the file's timestamp during backup.
6. Supports exclusion of files and directories.
7. It uses various standard utilities like Dump and GNU tar, which help to recover data very easily.
8. Its unique scheduler optimises the backup levels for